MLFlow on Azure ML

Dependencies

Step 1: Check if you have an Azure account Go to: https://portal.azure.com — sign in with your Microsoft account. If you don't have one, free tier gives you $200 credit for 30 days.

Step 2: Create a Resource Group In Azure Portal: Search bar → "Resource Groups" → Create Name: rg-mlops-demo Region: East US (or closest to you) Click: Review + Create → Create

Step 3: Create an Azure ML Workspace Search bar → "Azure Machine Learning" → Create Workspace name: mlops-workspace Resource group: rg-mlops-demo Region: East US Click: Review + Create → Create This takes 2-3 minutes.

Step 4: Install required packages cmdpip install azure-ai-ml azureml-mlflow azure-identity

```
#!pip install azure-ai-ml azureml-mlflow azure-identity
```

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Collecting azure-ai-ml
  Downloading azure_ai_ml-1.32.0-py3-none-any.whl.metadata (41 kB)
Collecting azureml-mlflow
  Downloading azureml_mlflow-1.62.0.post2-py3-none-any.whl.metadata
(2.9 kB)
Collecting azure-identity
  Downloading azure_identity-1.25.3-py3-none-any.whl.metadata (91 kB)
Requirement already satisfied: pyyaml<7.0.0,>=5.1.0 in C:\Users\gyanr\
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packages (from azure-ai-ml) (6.0.3)
Collecting azure-core>=1.23.0 (from azure-ai-ml)
  Downloading azure_core-1.39.0-py3-none-any.whl.metadata (48 kB)
Collecting azure-mgmt-core>=1.3.0 (from azure-ai-ml)
  Downloading azure_mgmt_core-1.6.0-py3-none-any.whl.metadata (4.6 kB)
Collecting marshmallow<4.0.0,>=3.5 (from azure-ai-ml)
  Using cached marshmallow-3.26.2-py3-none-any.whl.metadata (7.3 kB)
Requirement already satisfied: jsonschema<5.0.0,>=4.0.0 in C:\Users\
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Requirement already satisfied: tqdm<5.0.0 in C:\Users\gyanr\gyan-
python-workspace\bayesian-optimization\.venv_3.11\Lib\site-packages
(from azure-ai-ml) (4.67.1)
Collecting strictyaml<2.0.0 (from azure-ai-ml)
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Requirement already satisfied: colorama<1.0.0 in C:\Users\gyanr\gyan-
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(from azure-ai-ml) (0.4.6)
Collecting pyjwt<3.0.0 (from azure-ai-ml)
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Collecting azure-storage-file-share (from azure-ai-ml)
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any.whl.metadata (53 kB)
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Collecting pydash<9.0.0,>=6.0.0 (from azure-ai-ml)
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Requirement already satisfied: typing-extensions<5.0.0 in C:\Users\
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(from jsonschema<5.0.0,>=4.0.0->azure-ai-ml) (25.4.0)
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C:\Users\gyanr\gyan-python-workspace\bayesian-optimization\.venv_3.11\
Lib\site-packages (from jsonschema<5.0.0,>=4.0.0->azure-ai-ml)
(2025.9.1)
Requirement already satisfied: referencing>=0.28.4 in C:\Users\gyanr\
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packages (from jsonschema<5.0.0,>=4.0.0->azure-ai-ml) (0.37.0)
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Requirement already satisfied: packaging>=17.0 in C:\Users\gyanr\gyan-
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(from marshmallow<4.0.0,>=3.5->azure-ai-ml) (25.0)
Requirement already satisfied: python-dateutil>=2.6.0 in C:\Users\
gyanr\gyan-python-workspace\bayesian-optimization\.venv_3.11\Lib\site-
packages (from strictyaml<2.0.0->azure-ai-ml) (2.9.0.post0)
Collecting jsonpickle<5.0.0 (from azureml-mlflow)
  Downloading jsonpickle-4.1.1-py3-none-any.whl.metadata (8.1 kB)
Requirement already satisfied: mlflow-skinny<=3.9.0 in C:\Users\gyanr\
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packages (from azureml-mlflow) (3.8.1)
Collecting msrest>=0.6.18 (from azureml-mlflow)
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Collecting azure-storage-blob>=12.10.0 (from azure-ai-ml)
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Requirement already satisfied: pytz<=2025.2 in C:\Users\gyanr\gyan-
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(from azureml-mlflow) (2025.2)
Requirement already satisfied: requests>=2.21.0 in C:\Users\gyanr\
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Requirement already satisfied: cachetools<7,>=5.0.0 in C:\Users\gyanr\
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Requirement already satisfied: click<9,>=7.0 in C:\Users\gyanr\gyan-
python-workspace\bayesian-optimization\.venv_3.11\Lib\site-packages
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Requirement already satisfied: cloudpickle<4 in C:\Users\gyanr\gyan-
python-workspace\bayesian-optimization\.venv_3.11\Lib\site-packages
(from mlflow-skinny<=3.9.0->azureml-mlflow) (3.1.2)
Requirement already satisfied: databricks-sdk<1,>=0.20.0 in C:\Users\
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Requirement already satisfied: fastapi<1 in C:\Users\gyanr\gyan-
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Requirement already satisfied: gitpython<4,>=3.1.9 in C:\Users\gyanr\
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C:\Users\gyanr\gyan-python-workspace\bayesian-optimization\.venv_3.11\
Lib\site-packages (from mlflow-skinny<=3.9.0->azureml-mlflow) (8.7.1)
Requirement already satisfied: opentelemetry-api<3,>=1.9.0 in C:\
Users\gyanr\gyan-python-workspace\bayesian-optimization\.venv_3.11\
Lib\site-packages (from mlflow-skinny<=3.9.0->azureml-mlflow) (1.39.1)
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Users\gyanr\gyan-python-workspace\bayesian-optimization\.venv_3.11\
Lib\site-packages (from mlflow-skinny<=3.9.0->azureml-mlflow) (1.39.1)
Requirement already satisfied: opentelemetry-sdk<3,>=1.9.0 in C:\
Users\gyanr\gyan-python-workspace\bayesian-optimization\.venv_3.11\
Lib\site-packages (from mlflow-skinny<=3.9.0->azureml-mlflow) (1.39.1)
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Requirement already satisfied: pydantic<3,>=2.0.0 in C:\Users\gyanr\
gyan-python-workspace\bayesian-optimization\.venv_3.11\Lib\site-
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Requirement already satisfied: google-auth~=2.0 in C:\Users\gyanr\
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>azureml-mlflow) (2.47.0)
Requirement already satisfied: starlette<0.51.0,>=0.40.0 in C:\Users\
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Requirement already satisfied: rsa<5,>=3.1.4 in C:\Users\gyanr\gyan-
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skinny<=3.9.0->azureml-mlflow) (4.9.1)
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sdk<3,>=1.9.0->mlflow-skinny<=3.9.0->azureml-mlflow) (0.60b1)
Requirement already satisfied: annotated-types>=0.6.0 in C:\Users\
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mlflow) (2.41.5)
Requirement already satisfied: typing-inspection>=0.4.2 in C:\Users\
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mlflow) (0.4.2)
Requirement already satisfied: six>=1.5 in C:\Users\gyanr\gyan-python-
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(2.6.3)
Requirement already satisfied: certifi>=2017.4.17 in C:\Users\gyanr\
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(2026.1.4)
Requirement already satisfied: pyasn1>=0.1.3 in C:\Users\gyanr\gyan-
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>mlflow-skinny<=3.9.0->azureml-mlflow) (0.6.2)
Requirement already satisfied: anyio<5,>=3.6.2 in C:\Users\gyanr\gyan-
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>azureml-mlflow) (4.12.1)
Requirement already satisfied: h11>=0.8 in C:\Users\gyanr\gyan-python-
workspace\bayesian-optimization\.venv_3.11\Lib\site-packages (from
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Collecting msal-extensions>=1.2.0 (from azure-identity)
  Downloading msal_extensions-1.3.1-py3-none-any.whl.metadata (7.8 kB)
INFO: pip is looking at multiple versions of azure-storage-file-
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requirements. This could take a while.
Collecting azure-storage-file-datalake>=12.2.0 (from azure-ai-ml)
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Requirement already satisfied: pycparser in C:\Users\gyanr\gyan-
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python-workspace\bayesian-optimization\.venv_3.11\Lib\site-packages
(from cffi>=2.0.0->cryptography<47.0.0->azureml-mlflow) (2.23)
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(11 kB)
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>msrest>=0.6.18->azureml-mlflow)
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>azureml-mlflow)
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any.whl.metadata (2.2 kB)
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monitor-opentelemetry->azure-ai-ml)
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monitor-opentelemetry->azure-ai-ml)
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any.whl.metadata (3.5 kB)
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monitor-opentelemetry->azure-ai-ml)
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any.whl.metadata (1.9 kB)
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azure-monitor-opentelemetry->azure-ai-ml)
  Downloading opentelemetry_resource_detector_azure-0.1.5-py3-none-
any.whl.metadata (5.3 kB)
INFO: pip is looking at multiple versions of opentelemetry-sdk to
determine which version is compatible with other requirements. This
could take a while.
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  Downloading azure_monitor_opentelemetry-1.8.6-py3-none-
any.whl.metadata (25 kB)
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>azureml-mlflow)
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any.whl.metadata (2.2 kB)
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monitor-opentelemetry->azure-ai-ml)
  Downloading opentelemetry_instrumentation_flask-0.60b0-py3-none-
any.whl.metadata (2.2 kB)
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any.whl.metadata (2.1 kB)
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monitor-opentelemetry->azure-ai-ml)
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any.whl.metadata (2.6 kB)
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monitor-opentelemetry->azure-ai-ml)
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any.whl.metadata (3.4 kB)
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monitor-opentelemetry->azure-ai-ml)
  Downloading opentelemetry_instrumentation_urllib3-0.60b0-py3-none-
any.whl.metadata (4.2 kB)
Collecting azure-monitor-opentelemetry (from azure-ai-ml)
  Downloading azure_monitor_opentelemetry-1.8.5-py3-none-
any.whl.metadata (24 kB)
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  Downloading azure_monitor_opentelemetry-1.8.4-py3-none-
any.whl.metadata (24 kB)
  Downloading azure_monitor_opentelemetry-1.8.3-py3-none-
any.whl.metadata (23 kB)
  Downloading azure_monitor_opentelemetry-1.8.2-py3-none-
any.whl.metadata (23 kB)
Collecting opentelemetry-instrumentation-django~=0.57b0 (from azure-
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any.whl.metadata (2.3 kB)
Collecting opentelemetry-instrumentation-fastapi~=0.57b0 (from azure-
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Collecting opentelemetry-instrumentation-urllib3~=0.57b0 (from azure-
monitor-opentelemetry->azure-ai-ml)
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INFO: pip is looking at multiple versions of azure-monitor-
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Collecting azure-monitor-opentelemetry-exporter~=1.0.0b43 (from azure-
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  Downloading azure_monitor_opentelemetry_exporter-1.0.0b45-py2.py3-
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Requirement already satisfied: psutil<8,>=5.9 in C:\Users\gyanr\gyan-
python-workspace\bayesian-optimization\.venv_3.11\Lib\site-packages
(from azure-monitor-opentelemetry-exporter~=1.0.0b43->azure-monitor-
opentelemetry->azure-ai-ml) (7.2.1)
Collecting opentelemetry-instrumentation-wsgi==0.62b0 (from
opentelemetry-instrumentation-django~=0.57b0->azure-monitor-
opentelemetry->azure-ai-ml)
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any.whl.metadata (2.2 kB)
Collecting opentelemetry-instrumentation==0.62b0 (from opentelemetry-
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INFO: pip is looking at multiple versions of opentelemetry-
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Collecting opentelemetry-instrumentation-wsgi==0.61b0 (from
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Collecting opentelemetry-util-http==0.60b1 (from opentelemetry-
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(2.6 kB)
Collecting wrapt<2.0.0,>=1.0.0 (from opentelemetry-
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instrumentation==0.60b1->opentelemetry-instrumentation-django~=0.57b0-
>azure-monitor-opentelemetry->azure-ai-ml)
  Downloading wrapt-1.17.3-cp311-cp311-win_amd64.whl.metadata (6.5 kB)
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Collecting opentelemetry-instrumentation-fastapi~=0.57b0 (from azure-
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Collecting asgiref~=3.0 (from opentelemetry-instrumentation-
asgi==0.60b1->opentelemetry-instrumentation-fastapi~=0.57b0->azure-
monitor-opentelemetry->azure-ai-ml)
  Downloading asgiref-3.11.1-py3-none-any.whl.metadata (9.3 kB)
INFO: pip is looking at multiple versions of opentelemetry-
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other requirements. This could take a while.
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Collecting opentelemetry-instrumentation-dbapi==0.62b0 (from
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Collecting opentelemetry-instrumentation-psycopg2~=0.57b0 (from azure-
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Collecting opentelemetry-instrumentation-dbapi==0.60b1 (from
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INFO: pip is looking at multiple versions of opentelemetry-
instrumentation-requests to determine which version is compatible with
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INFO: pip is looking at multiple versions of opentelemetry-
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other requirements. This could take a while.
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monitor-opentelemetry->azure-ai-ml)
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any.whl.metadata (3.4 kB)
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instrumentation-urllib3 to determine which version is compatible with
other requirements. This could take a while.
Collecting opentelemetry-instrumentation-urllib3~=0.57b0 (from azure-
monitor-opentelemetry->azure-ai-ml)
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any.whl.metadata (4.2 kB)
Downloading azure_ai_ml-1.32.0-py3-none-any.whl (13.2 MB)
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   ------------------ --------------------- 6.0/13.2 MB 33.5 MB/s eta
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   ---------------------------------------  13.1/13.2 MB 39.0 MB/s eta
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   ---------------------------------------- 13.2/13.2 MB 22.9 MB/s
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Using cached isodate-0.7.2-py3-none-any.whl (22 kB)
Using cached marshmallow-3.26.2-py3-none-any.whl (50 kB)
Downloading pydash-8.0.6-py3-none-any.whl (101 kB)
Downloading pyjwt-2.12.1-py3-none-any.whl (29 kB)
Downloading strictyaml-1.7.3-py3-none-any.whl (123 kB)
Downloading azureml_mlflow-1.62.0.post2-py3-none-any.whl (1.0 MB)
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Downloading azure_core-1.39.0-py3-none-any.whl (218 kB)
Downloading azure_mgmt_core-1.6.0-py3-none-any.whl (29 kB)
Downloading azure_storage_blob-12.27.1-py3-none-any.whl (428 kB)
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Downloading jsonpickle-4.1.1-py3-none-any.whl (47 kB)
Downloading azure_identity-1.25.3-py3-none-any.whl (192 kB)
Downloading azure_common-1.1.28-py2.py3-none-any.whl (14 kB)
Downloading azure_storage_file_datalake-12.22.0-py3-none-any.whl (264
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Downloading msal-1.36.0-py3-none-any.whl (121 kB)
Downloading msal_extensions-1.3.1-py3-none-any.whl (20 kB)
Downloading msrest-0.7.1-py3-none-any.whl (85 kB)
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Downloading opentelemetry_instrumentation-0.60b1-py3-none-any.whl (33
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Downloading opentelemetry_instrumentation_wsgi-0.60b1-py3-none-any.whl
(14 kB)
Downloading opentelemetry_util_http-0.60b1-py3-none-any.whl (8.9 kB)
Downloading opentelemetry_instrumentation_fastapi-0.60b1-py3-none-
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Downloading opentelemetry_instrumentation_asgi-0.60b1-py3-none-any.whl
(16 kB)
Downloading asgiref-3.11.1-py3-none-any.whl (24 kB)
Downloading opentelemetry_instrumentation_flask-0.60b1-py3-none-
any.whl (15 kB)
Downloading opentelemetry_instrumentation_psycopg2-0.60b1-py3-none-
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any.whl (13 kB)
Downloading opentelemetry_instrumentation_requests-0.60b1-py3-none-
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Downloading opentelemetry_instrumentation_urllib-0.60b1-py3-none-
any.whl (12 kB)
Downloading opentelemetry_instrumentation_urllib3-0.60b1-py3-none-
any.whl (13 kB)
Downloading opentelemetry_resource_detector_azure-0.1.5-py3-none-
any.whl (14 kB)
Downloading wrapt-1.17.3-cp311-cp311-win_amd64.whl (38 kB)
Downloading azure_storage_file_share-12.24.0-py3-none-any.whl (314 kB)
Installing collected packages: azure-common, wrapt, pyjwt, pydash,
opentelemetry-util-http, oauthlib, marshmallow, jsonpickle, isodate,
asgiref, strictyaml, requests-oauthlib, azure-core, msrest, azure-
storage-file-share, azure-storage-blob, azure-mgmt-core, azure-core-
tracing-opentelemetry, opentelemetry-instrumentation, msal, azure-
storage-file-datalake, opentelemetry-resource-detector-azure,
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opentelemetry-instrumentation-asgi, msal-extensions, opentelemetry-
instrumentation-psycopg2, opentelemetry-instrumentation-flask,
opentelemetry-instrumentation-fastapi, opentelemetry-instrumentation-
django, azure-identity, azureml-mlflow, azure-monitor-opentelemetry-
exporter, azure-monitor-opentelemetry, azure-ai-ml

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Successfully installed asgiref-3.11.1 azure-ai-ml-1.32.0 azure-common-
1.1.28 azure-core-1.39.0 azure-core-tracing-opentelemetry-1.0.0b12
azure-identity-1.25.3 azure-mgmt-core-1.6.0 azure-monitor-
opentelemetry-1.8.2 azure-monitor-opentelemetry-exporter-1.0.0b45
azure-storage-blob-12.27.1 azure-storage-file-datalake-12.22.0 azure-
storage-file-share-12.24.0 azureml-mlflow-1.62.0.post2 isodate-0.7.2
jsonpickle-4.1.1 marshmallow-3.26.2 msal-1.36.0 msal-extensions-1.3.1
msrest-0.7.1 oauthlib-3.3.1 opentelemetry-instrumentation-0.60b1
opentelemetry-instrumentation-asgi-0.60b1 opentelemetry-
instrumentation-dbapi-0.60b1 opentelemetry-instrumentation-django-
0.60b1 opentelemetry-instrumentation-fastapi-0.60b1 opentelemetry-
instrumentation-flask-0.60b1 opentelemetry-instrumentation-psycopg2-
0.60b1 opentelemetry-instrumentation-requests-0.60b1 opentelemetry-
instrumentation-urllib-0.60b1 opentelemetry-instrumentation-urllib3-
0.60b1 opentelemetry-instrumentation-wsgi-0.60b1 opentelemetry-
resource-detector-azure-0.1.5 opentelemetry-util-http-0.60b1 pydash-
8.0.6 pyjwt-2.12.1 requests-oauthlib-2.0.0 strictyaml-1.7.3 wrapt-
1.17.3
```

Step 5: Get your MLFlow tracking URI In Azure Portal: Go to your ML Workspace Click "Overview" Copy "MLflow tracking URI" Looks like: azureml://eastus.api.azureml.ms/mlflow/v1.0/subscriptions/.../resourceGroups/.../providers/Microsoft.MachineLearningServices/workspaces/mlops-workspace

Setup

```
mlflow="azureml://eastus.api.azureml.ms/mlflow/v1.0/subscriptions/
5395efdb-f3cd-4215-9872-be0fe68fca37/resourceGroups/rg-mlops-demo/
providers/Microsoft.MachineLearningServices/workspaces/mlops-ws-
mlflow-demo"
```

title

What the notebook will log to Azure MLFlow per request:

Params:

- model
- temperature
- top_p
- frequency_penalty
- presence_penalty

Metrics:

- request_latency
- prompt_tokens
- completion_tokens
- conversation_tokens
- request_count

Tiktoken

```
import os
def _set_env_from_file(var: str, file_path: str = r"C:\Users\gyanr\
gyan-python-workspace\grk_langchain\app\notebooks\data\
openai_key.txt"):
    """
    Reads an API key from a specified file and sets it as an
environment variable.
    """
    if not os.environ.get(var):
        try:
            # The 'with open' statement ensures the file is closed
automatically
            with open(file_path, 'r') as f:
                # Read the first line and strip any leading/trailing
whitespace
                key = f.readline().strip()

            if key:
                os.environ[var] = key
                print(f"Successfully loaded {var} from {file_path}")
```

```
            else:
                print(f"Warning: {file_path} is empty.")

        except FileNotFoundError:
            print(f"Error: Key file not found at {file_path}. Please
create the file.")


from openai import OpenAI

open_api_var='OPENAI_API_KEY'
_set_env_from_file(open_api_var,file_path=r"C:\Users\gyanr\gyan-
python-workspace\generative-ai\resources\keys.txt")
print(os.environ.get('OPENAI_API_KEY'))
```

```
Successfully loaded OPENAI_API_KEY from C:\Users\gyanr\gyan-python-
workspace\generative-ai\resources\keys.txt
sk-proj-lW2GI83bGmvoehMvXuCUGFw5snuY0ByY6vVTwZcRLPI15-46Q8Ku5-
RpNs409bV0zA4UR3iL-WT3BlbkFJ7ddIvVSWWTzsbROr7R0t-Z1IqsZj-
jvbHLopzmv8uB2SekCa5p1dyqmrNkPURQUf-GAGqiJ4IA
```

```
import os
import time
import random
import mlflow
import tiktoken as tk
from openai import OpenAI
from azure.ai.ml import MLClient
from azure.identity import InteractiveBrowserCredential
from colorama import Fore, Style, init

# Initialize colorama
init()

# Constants
MODEL = "gpt-3.5-turbo"
TEMPERATURE = 0.7
TOP_P = 1
FREQUENCY_PENALTY = 0
PRESENCE_PENALTY = 0
MAX_TOKENS = 800
DEBUG = False

# OpenAI client
client = OpenAI(api_key=os.getenv("OPENAI_API_KEY"))

print("Imports and configuration complete")
print(f"Model:       {MODEL}")
print(f"Temperature: {TEMPERATURE}")
print(f"Max tokens:  {MAX_TOKENS}")
```

```
Imports and configuration complete
Model:       gpt-3.5-turbo
Temperature: 0.7
Max tokens:  800
```

```python
from azure.identity import InteractiveBrowserCredential
from azure.ai.ml import MLClient
import mlflow


credential = InteractiveBrowserCredential()

ml_client = MLClient(
    credential=credential,
    subscription_id="5395efdb-f3cd-4215-9872-be0fe68fca37",      #
from Azure Portal
    resource_group_name="rg-mlops-demo",
    workspace_name="mlops-ws-mlflow-demo"
)

# get MLFlow tracking URI
mlflow_tracking_uri = ml_client.workspaces.get(
    ml_client.workspace_name
).mlflow_tracking_uri

print(mlflow_tracking_uri)

# set it
mlflow.set_tracking_uri(mlflow_tracking_uri)
mlflow.set_experiment("genai_mlflow_demo")
```

```
Class DeploymentTemplateOperations: This is an experimental class, and
may change at any time. Please see https://aka.ms/azuremlexperimental
for more information.
C:\Users\gyanr\gyan-python-workspace\bayesian-optimization\.venv_3.11\
Lib\site-packages\msal\oauth2cli\oauth2.py:408: UserWarning:
response_mode='form_post' is recommended for better security. See
https://www.rfc-editor.org/rfc/rfc9700.html#section-4.3.1
  warnings.warn(

azureml://eastus.api.azureml.ms/mlflow/v2.0/subscriptions/5395efdb-
f3cd-4215-9872-be0fe68fca37/resourceGroups/rg-mlops-demo/providers/
Microsoft.MachineLearningServices/workspaces/mlops-ws-mlflow-demo

<Experiment: artifact_location='', creation_time=1775925360525,
experiment_id='e86b4355-5f3a-4917-a494-b8ecab7b0514',
last_update_time=None, lifecycle_stage='active',
name='genai_mlflow_demo', tags={}>
```

```python
print("Connected to Azure MLFlow successfully")
print(f"Tracking URI: {mlflow_tracking_uri}")
```

```
Connected to Azure MLFlow successfully
Tracking URI:
azureml://eastus.api.azureml.ms/mlflow/v2.0/subscriptions/5395efdb-
f3cd-4215-9872-be0fe68fca37/resourceGroups/rg-mlops-demo/providers/
Microsoft.MachineLearningServices/workspaces/mlops-ws-mlflow-demo
```

title

```
def count_tokens(string: str, encoding_name: str = "cl100k_base") ->
int:
    """
    Count the number of tokens in a string using tiktoken.

    Args:
        string: text to count tokens for
        encoding_name: encoding to use
                       cl100k_base is used by gpt-3.5-turbo and gpt-4
    Returns:
        number of tokens
    """
    encoding = tk.get_encoding(encoding_name)
    num_tokens = len(encoding.encode(string, disallowed_special=()))
    return num_tokens


# Test with examples
examples = [
    "Can you answer medical question?",
    "Is ADHD that serious?",
    "Tell me a detailed story about a boy who became successful with
god's grace",
    "What is the meaning of life, the universe, and everything in it?"
]

print(f"{'Text':<65} {'Tokens':>6}")
print("-" * 72)
for ex in examples:
    print(f"{ex:<65} {count_tokens(ex):>6}")
```

```
Text
Tokens
------------------------------------------------------------------------
--
Can you answer medical question?
6
Is ADHD that serious?
5
Tell me a detailed story about a boy who became successful with god's
grace     15
```

```
What is the meaning of life, the universe, and everything in it?
15

def generate_text(conversation: list, max_tokens: int = MAX_TOKENS) ->
str:
    """
    Call OpenAI API, measure latency, count tokens,
    and log all metrics to Azure MLFlow.

    Args:
        conversation: list of message dicts with role and content
        max_tokens: maximum tokens in the response

    Returns:
        AI response as string
    """
    start_time = time.time()

    response = client.chat.completions.create(
        model=MODEL,
        messages=conversation,
        temperature=TEMPERATURE,
        max_tokens=max_tokens,
        top_p=TOP_P,
        frequency_penalty=FREQUENCY_PENALTY,
        presence_penalty=PRESENCE_PENALTY
    )

    latency = time.time() - start_time
    message_response = response.choices[0].message.content

    # Count tokens
    prompt_tokens       = count_tokens(conversation[-1]['content'])
    conversation_tokens = count_tokens(str(conversation))
    completion_tokens   = count_tokens(message_response)

    if DEBUG:
        run = mlflow.active_run()
        print(f"Run ID: {run.info.run_id}")

    # Log metrics to Azure MLFlow
    mlflow.log_metrics({
        "request_latency":     latency,
        "prompt_tokens":       prompt_tokens,
        "completion_tokens":   completion_tokens,
        "conversation_tokens": conversation_tokens,
        "request_count":       1
    })

    # Log parameters to Azure MLFlow
```

```
    mlflow.log_params({
        "model":             MODEL,
        "temperature":       TEMPERATURE,
        "top_p":             TOP_P,
        "frequency_penalty": FREQUENCY_PENALTY,
        "presence_penalty":  PRESENCE_PENALTY,
        "max_tokens":        max_tokens
    })

    return message_response


print("generate_text() defined successfully")
```

```
generate_text() defined successfully
```

```
# Test prompts — mirrors wonderwords auto-generation approach
test_inputs = [
    "Hello, how are you?",
    "What is the capital of France?",
    "Tell me a dad joke",
    "Tell me a short story",
    "What is the meaning of life?",
    "What is the largest mammal?",
    "What is the square root of 144?",
    "Give me a productivity tip",
    "Explain quantum computing in one sentence",
    "What is machine learning?"
]

LOAD_TEST_ITERATIONS = 10

print(f"Starting load test — {LOAD_TEST_ITERATIONS} iterations")
print("=" * 60)

with mlflow.start_run():
    conversation = [
        {"role": "system", "content": "You are a helpful assistant."},
    ]

    for i in range(LOAD_TEST_ITERATIONS):
        # randomly select a prompt
        user_input = random.choice(test_inputs)
        conversation.append({"role": "user", "content": user_input})

        # generate response and log to MLFlow
        ai_output = generate_text(conversation, MAX_TOKENS)
        conversation.append({"role": "assistant", "content":
ai_output})

        print(f"\nIteration {i+1}/{LOAD_TEST_ITERATIONS}")
```

```
        print(f"{Fore.GREEN}User:{Style.RESET_ALL} {user_input}")
        print(f"{Fore.BLUE}AI:  {Style.RESET_ALL}
{ai_output[:100]}...")

print("\n" + "=" * 60)
print("Load test complete — check Azure ML Studio for metrics")
```

```
Starting load test — 10 iterations
============================================================

Iteration 1/10
User: Give me a productivity tip
AI:   One helpful productivity tip is to break down your tasks into
smaller, more manageable chunks. This ...

Iteration 2/10
User: Give me a productivity tip
AI:   Another productivity tip is to eliminate distractions while
working. This can include turning off no...

Iteration 3/10
User: Explain quantum computing in one sentence
AI:   Quantum computing is a type of computing that uses quantum-
mechanical phenomena, such as superpositi...

Iteration 4/10
User: What is the capital of France?
AI:   The capital of France is Paris....

Iteration 5/10
User: What is the meaning of life?
AI:   The meaning of life is a philosophical question that has been
pondered by humans for centuries, and ...

Iteration 6/10
User: What is the largest mammal?
AI:   The largest mammal is the blue whale, which is also the largest
animal on Earth....

Iteration 7/10
User: Hello, how are you?
AI:   Hello! I'm here and ready to help you. How can I assist you
today?...

Iteration 8/10
User: What is the meaning of life?
AI:   The meaning of life is a profound and subjective question that
can have different interpretations fo...

Iteration 9/10
```

```
User: What is machine learning?
AI:   Machine learning is a subset of artificial intelligence that
involves developing algorithms and stat...

Iteration 10/10
User: What is the capital of France?
AI:   The capital of France is Paris....
 View run placid_brain_28z3qb3d at:
https://eastus.api.azureml.ms/mlflow/v2.0/subscriptions/5395efdb-f3cd-
4215-9872-be0fe68fca37/resourceGroups/rg-mlops-demo/providers/
Microsoft.MachineLearningServices/workspaces/mlops-ws-mlflow-demo/#/
experiments/e86b4355-5f3a-4917-a494-b8ecab7b0514/runs/5d35429f-1d05-
4602-a5ca-3880dc68929f
 View experiment at:
https://eastus.api.azureml.ms/mlflow/v2.0/subscriptions/5395efdb-f3cd-
4215-9872-be0fe68fca37/resourceGroups/rg-mlops-demo/providers/
Microsoft.MachineLearningServices/workspaces/mlops-ws-mlflow-demo/#/
experiments/e86b4355-5f3a-4917-a494-b8ecab7b0514

============================================================
Load test complete — check Azure ML Studio for metrics
```

title

title

```

```

Conclusion